

Pretrained Patient Trajectories for Adverse Drug Event Prediction Using Common Data Model-based Electronic Health Records

Corresponding Author: Professor Kwangsoo Kim

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In this work, the authors develop EHR pertaining models based on BERT and proceed by fine-tuning them in order to adapt them to the task of predicting adverse drug events. The work is completed with an analysis of the importance of features that allows for a better interpretability of the results. I believe that this is a strong aspect that improves the overall quality of the work.

The paper is very well written and addresses an interesting problem in the medical field, which is the automatic recognition of the development of adverse events due to the assumption of drugs.

The introduction is complete and does a good job of introducing the topic of the paper and contextualizing the author's work in the field.

The methods section is also very well made, and the authors here describe how the dataset needed to train the models was built (including how the outcome was defined) and how the models were adjusted to fit the task of interest. This aspect improves the work reproducibility. The authors also provide the source code of the experiments, another great way to allow others to replicate their results or try these approaches on different datasets.

Finally, the results are very well presented and discussed in the results and discussion section.

I believe the quality of this work to be high and I have no further comments.

Reviewer #2

(Remarks to the Author)

Dear authors,

Firstly, I want to thank you for the opportunity to review your manuscript. Your research was evidently performed at the highest level of quality, and I believe that this study on the performance of transformer models for predicting ADE events is of high clinical interest. I commend you for your effort, and I only have a few suggestions to improve the manuscript's reader comprehension:

Methods

Record Tokenization

'Regarding special tokens, we utilized five special tokens: [PAD], [MASK], [UNK], [CLS], and [SEP]'

As you have brought attention to the special tokens and explained the [SEP] token, it would be beneficial to briefly explain the role of the other 4 tokens (e.g. [UNK] represents an 'out-of-vocabulary' token)

Pretraining Process

'The models were pretrained with 100 epochs, and we selected the model with the minimum CE loss during the training'

The CE acronym should be defined here, as I haven't found a definition before this instance. (i.e. [...] Cross-Entropy (CE) Loss [...])

'For type I, we used the model pretrained with the internal dataset without additional pretraining. For type II, we used the external dataset to additionally pretrain the model initially pretrained by the internal dataset. For type III, we initially pretrained the initial model using the external dataset. The type II and III models were pretrained using the external dataset for ten epochs.'

Reading this section and referencing Figure 1, I am not sure I correctly understood the pretraining routine for each type. Please find below a rephrasing suggestion, under my understanding of the routines.

For type I, we pretrained the model on just the internal dataset (Hospital A) for 100 epochs. For type II, we pretrained the model on the internal dataset (Hospital A), followed by an additional 10 epochs of pretraining on the external dataset (Hospital B). For type III, we only pretrained the model on the external dataset (Hospital B) for 10 epochs.'

If I misunderstood the pretraining routines, please feel free to edit the text as you see fit.

Results

Performance Evaluation

Figure 2 shows some impressive results for the models. I would've preferred that the figure was split into two figures stacked vertically, Figure 2.a. Internal validation, Figure 2.b. External Validation, so that I could better examine the results. This suggestion is a matter of personal preference and a desire to put more emphasis on the external validation results.

Discussion

'Given that only 7,000 codes were common between the two hospitals, which had 26,417 and 22,551 codes, respectively, it is noteworthy that the model pretrained with numerous unknown codes remained effective'

The statement above is very salient to contextualize the differences in results observed between internal and external validation. I would recommend adding a column showing the intersection of the tokens between Internal and External Datasets in Supplementary Table 3.

Reviewer #3

(Remarks to the Author)

The article introduced an ADE prediction method using a pre-trained language model with tokenized EHR data. The proposed method used the BERT model and combined condition, measurement, drug, and procedure with domain labels to enhance the prediction ability of the trained model. The model was able to learn series modalities of the EHR trajectory with domain embedding, and predict ADE occurrence of certain drugs. As mentioned by the author, the proposed method is a pioneer study of ADE prediction tasks using an EHR pre-trained model. The result indicated good AUC performance to warn potential ADE of the patient. And the author validated that the internal pre-trained model with external pretraining outperformed other settings. Although a BERT-based model using tokenized EHR data has already been reported in several studies, the proposed method considered domain embedding, numerical embedding, and internal-external pretraining to modify the technique to suit ADE scenarios better. It may help extend the application domain of the EHR pre-trained model in serial problems.

There are some questions I would like to communicate with the author:

(1) In the Introduction Section, the OMOP-CDM is mentioned to emphasize the importance of cross-hospital data analysis. Although it's important to perform multicenter data training and validation, the first paragraph seems weakly related to the major claims of the study. What's the importance of EHR data and OMOP CDM for ADE prediction problems? The structure of the Introduction may need reform.

(2) The review of current studies is insufficient. There are several BERT-based and transformer-based EHR pre-trained model functions in clinical domains. Reviewing the current studies and pointing out the empty in ADE prediction would be better. Also, studies in ADE prediction using other methods are preferred to summarize their disadvantages.

(3) The prediction task set a timepoint 4 weeks before ADE as the cohort requires an ADE within 4 weeks of prescription. As illustrated in Fig 1b, the target drug and related ADE may both occur after the prediction timepoint. How to explain the prediction task without a prescription at the time of prediction timepoint? Since 2,4,8,12 weeks' timepoint may cause the prescription to occur before or after the prediction timepoint, will the task setting influence the performance of the validation of different prediction timepoints?

(4) Fig 3 and Fig 4 showed the feature importance of 3 ADEs. As there are 7000/41536 common concepts in both hospitals, what are the common features and unique features of the top 10s?

(5) Table 2 indicated an unbalanced data distribution of the cohort where the control group is much larger than the case group. Still, the model presented a good recall. There should be a comparatively large false positive, though it is acceptable to catch any possible ADE. It is better to report the percentage of case and control in the study cohort characteristics. Discussing examples of false positives and negatives is preferred to better explain the validation result.

Reviewer #4

(Remarks to the Author)

This is an interesting and well written paper. The clinical aspect describes a proof of concept for predicting drug associated adverse events. There is a clear clinical rationale for this and the concept is clinically valid - this technique could be applied to a precision medicine approach to clinical decision support around drug choices in the future.

I would like to see more discussion around some of the features and what their relevance is for future implementation work. Many of the features are secondary to known risk factors for these adverse events (ie as you describe ISMN use would strongly suggest the patient has CVS disease, speech audiometry likely to be strongly correlated with advanced age). If the model has simply identified known and well established risk factors that a clinician uses in their assessment how would this add benefit in a clinical workflow? This will not be immediately obvious to clinicians reading the paper at present.

The paper would be further enriched by expanding on the limitations - particularly the fact that the work was completed on two large, well constructed and OMOP mapped datasets, with high quality EHR data, which is not the case in a large percentage of the world, limiting external validity. Further validation work on external datasets is essential (I appreciate this is likely to be in a future paper)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Reviewer #2

(Remarks to the Author)

Dear authors,

Thank you for taking the time to integrate the recommendations into your manuscript. I believe that the presentation of your already strong research has been improved through the text edits. I am looking forward to reading your future studies.

Reviewer #3

(Remarks to the Author)

The author has addressed my issues.

Reviewer #4

(Remarks to the Author)

I commend the authors for their rapid, comprehensive amendments to the manuscript and would recommend publication

Open Access This Peer Review File is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made.

In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.

The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>

21 January 2025

Editorial Board Member

RE: Response to review of paper (COMMSMED-24-1520), entitled "Pretrained Patient Trajectories for Adverse Drug Event Prediction Using Common Data Model-based Electronic Health Records"

The attached letter and manuscript include a complete point-by-point response to the concerns raised by the reviewers. These appeals are largely focused on addressing and defending the significant concern of the reviewers that the writing needed to be much clearer, based on the reviewers' comments.

We would be happy to address any further concerns regarding this manuscript. We appreciate your continued consideration.

Sincerely,

Kwangsoo Kim, PhD

Department of Transdisciplinary Medicine, Institute of Convergence Medicine with Innovative Technology, Seoul National University Hospital

Department of Medicine, College of Medicine, Seoul National University

Address: 101, Daehak-ro, Jongno-gu, Seoul, Republic of Korea

Tel: +82-2-2072-4489

Fax: +82-2-6072-5304

Email: kksoo716@gmail.com

To the Editor and Reviewers of submission (COMMSMED-24-1520):

We greatly appreciate the reviewers' constructive comments and the editor's decision regarding our manuscript. The reviewer expressed some concerns regarding the limitations of the original manuscript. We recognized the importance of advices and have therefore revised the manuscript extensively to the best of our ability.

Specific comments: Reviewer #1

Comment 1:

In this work, the authors develop EHR pertaining models based on BERT and proceed by fine-tuning them in order to adapt them to the task of predicting adverse drug events. The work is completed with an analysis of the importance of features that allows for a better interpretability of the results. I believe that this is a strong aspect that improves the overall quality of the work.

The paper is very well written and addresses an interesting problem in the medical field, which is the automatic recognition of the development of adverse events due to the assumption of drugs.

The introduction is complete and does a good job of introducing the topic of the paper and contextualizing the author's work in the field.

The methods section is also very well made, and the authors here describe how the dataset needed to train the models was built (including how the outcome was defined) and how the models were adjusted to fit the task of interest. This aspect improves the work reproducibility. The authors also provide the source code of the experiments, another great way to allow others to replicate their results or try these approaches on different datasets.

Finally, the results are very well presented and discussed in the results and discussion section.

I believe the quality of this work to be high and I have no further comments.

Answer:

Thank you for your thoughtful and positive comment on our manuscript. We greatly appreciate your recognition of our efforts to address the task of predicting adverse drug events using EHR pretraining models. Your kind words and encouragement motivate us to continue impactful research.

Specific comments: Reviewer #2

Comment 1:

Record Tokenization

‘Regarding special tokens, we utilized five special tokens: [PAD], [MASK], [UNK], [CLS], and [SEP]’

As you have brought attention to the special tokens and explained the [SEP] token, it would be beneficial to briefly explain the role of the other 4 tokens (e.g. [UNK] represents an ‘out-of-vocabulary’ token)

Answer:

Thank you for your valuable comments and we totally agree with your opinion. We have added explanations for the rest of the special tokens.

Changes made in the manuscript:

“Record tokenization” section in **Methods**.

[Before] (...) Like BERT used [SEP] to separate two sentences, we inserted [SEP] between all different dates. (...)

[Revised] (...) [PAD] was used to align patient trajectories with different lengths to the same lengths, and [MASK] was used for the MLM process. Tokens that were not in the training dataset were replaced with [UNK]. Every trajectory starts with [CLS] considering representation usage for various downstream tasks. Like BERT used [SEP] to separate two sentences, we inserted [SEP] between all different dates. (...)

Comment 2:

Pretraining Process

‘The models were pretrained with 100 epochs, and we selected the model with the minimum CE loss during the training’

The CE acronym should be defined here, as I haven’t found a definition before this instance. (i.e. [...] Cross-Entropy (CE) Loss [...])

Answer:

Thank you for your valuable comments. We have added the definition of CE.

Changes made in the manuscript:

The first paragraph of **“Pretraining process”** section in **Methods**.

[Before] (...) The models were pretrained with 100 epochs, and we selected the model with the minimum CE loss during the training.

[Revised] (...) The models were pretrained with 100 epochs, and we selected the model with the minimum cross-entropy (CE) loss during the training.

Comment 3:

‘For type I, we used the model pretrained with the internal dataset without additional pretraining. For type II, we used the external dataset to additionally pretrain the model initially pretrained by the internal dataset. For type III, we initially pretrained the initial model using the external dataset. The type II and III models were pretrained using the external dataset for ten epochs.’

Reading this section and referencing Fig. 1, I am not sure I correctly understood the pretraining routine for each type. Please find below a rephrasing suggestion, under my understanding of the routines.

For type I, we pretrained the model on just the internal dataset (Hospital A) for 100 epochs. For type II, we pretrained the model on the internal dataset (Hospital A), followed by an additional 10 epochs of pretraining on the external dataset (Hospital B). For type III, we only pretrained the model on the external dataset (Hospital B) for 10 epochs.’

If I misunderstood the pretraining routines, please feel free to edit the text as you see fit.

Answer:

Thank you for your valuable comments. You have understood our pretraining process correctly. However, we acknowledge that our description of the pretraining process might cause misunderstandings. Accordingly, we have revised our manuscript. Thank you again for your opinion, and we hope our revision will improve the readability.

Changes made in the manuscript:

The second paragraph of “**Pretraining process**” section in **Methods**.

[Before] For type I, we used the model pretrained with the internal dataset without additional pretraining. For type II, we used the external dataset to additionally pretrain the model initially pretrained by the internal dataset. For type III, we initially pretrained the initial model using the external dataset. The type II and III models were pretrained using the external dataset for ten epochs.

[Revised] For type I, we used the model initially pretrained with the internal dataset without additional pretraining. For type II, we additionally pretrained the model that was initially pretrained by the internal dataset using the external dataset. For type III, we initially pretrained the model only using the external dataset. The type II and III models were pretrained using the external dataset for ten epochs.

Comment 4:

Performance Evaluation in Results

Figure 2 shows some impressive results for the models. I would've preferred that the figure was split into two figures stacked vertically, Figure 2.a. Internal validation, Figure 2.b. External Validation, so that I could better examine the results. This suggestion is a matter of personal preference and a desire to put more emphasis on the stellar validation results.

Answer:

Thank you for your valuable comment. Even though the vertically stacked figure might be more intuitive, we are afraid that the figure can fit in the space of a paper. If you don't mind, please let us discuss this issue with the editor since this is an editing concern.

Comment 5:

'Given that only 7,000 codes were common between the two hospitals, which had 26,417 and 22,551 codes, respectively, it is noteworthy that the model pretrained with numerous unknown codes remained effective'

The statement above is very salient to contextualize the differences in results observed between internal and external validation. I would recommend adding a column showing the intersection of the tokens between Internal and External Datasets in Supplementary Table 3.

Answer:

Thank you for your helpful comment. We totally agree with your suggestion. We have added an extra column exhibiting the number of shared codes for each domain.

Changes made in the manuscript:**Supplementary Table 3.**

[Before]

Supplementary Table 3. Token counts for each domain in each dataset.

	Internal dataset	External dataset	Aggregated
Diagnosis	12,913	10,420	17,785
Prescription	6,152	6,332	12,213
Measurement	2,462	2,857	4,255
Procedure	4,883	2,935	7,276
Total	26,410 + 5 special tokens + 2 gender tokens	22,544 + 5 special tokens + 2 gender tokens	41529 + 5 special tokens + 2 gender tokens

[Revised]

Supplementary Table 3. Token counts for each domain in each dataset.

	Internal dataset	External dataset	Shared	Aggregated
Diagnosis	12,913	10,420	5,548	17,785
Prescription	6,152	6,332	271	12,213
Measurement	2,462	2,857	1,064	4,255
Procedure	4,883	2,935	542	7,276
Total	26,410 + 5 special tokens + 2 gender tokens	22,544 + 5 special tokens + 2 gender tokens	7,425 + 5 special tokens + 2 gender tokens	41,529 + 5 special tokens + 2 gender tokens

Specific comments: Reviewer #3

Comment 1:

In the Introduction Section, the OMOP-CDM is mentioned to emphasize the importance of cross-hospital data analysis. Although it's important to perform multicenter data training and validation, the first paragraph seems weakly related to the major claims of the study. What's the importance of EHR data and OMOP CDM for ADE prediction problems? The structure of the Introduction may need reform.

Answer:

Thank you for your valuable comment. The introduction consists of four parts: First, the explanation and global status of EHR data and OMOP CDM. Second, description of EHR pretraining models and their application. Third, the reason why ADE prediction using EHR pretraining models is promising. Fourth, the outline of our study.

The first paragraph was to attract readers' interest in EHR and OMOP CDM because we mainly utilized patient records from the OMOP CDM-based EHR database. However, as you indicated, combining the descriptions of EHR data and OMOP CDM in the same paragraph might be confusing. The importance of EHR data and OMOP CDM for ADE prediction problems can be explained separately; EHR contains a lot of patients' medical records. Especially for inpatients, dense records of drug, measurement, diagnosis, and procedure are contained, and those records can capture various drug reactions across different medications (described in the third paragraph). On the other hand, OMOP CDM facilitates multicenter studies and enhances the reliability of the study by standardizing EHR data.

We acknowledge that our introduction needs reform and complement. Accordingly, we have revised the manuscript to improve the readability. Once again, we appreciate your assistance in enhancing the manuscript.

Changes made in the manuscript:

The first paragraph of **Introduction**.

[Before] (...) The widespread use of EHR has increased the amount of healthcare data, and numerous studies have been conducted using those data for various medical tasks, such as disease prediction, patient status monitoring, and diagnosis support. (...)

[Revised] (...) The widespread use of EHR has increased the amount of healthcare data, and numerous machine learning studies have been conducted using EHR data for various medical tasks, such as disease prediction, patient status monitoring, and diagnosis support. (...)

Separation of first paragraph and change of paragraphs order of **Introduction**.

The first paragraph now ends with the sentence “(...) such as disease prediction, patient status monitoring, and diagnosis support.” We combined the following sentences that start with “Early works tried to use data from multiple hospitals;” with the last paragraph that summarizing our study. We emphasized the importance of cross-hospital data analysis through OMOP CDM-based EHR data. (Please refer to the changes made in the last paragraph.)

The third paragraph of **Introduction**.

[Before] EHR pretraining models exhibited great performance in various medical tasks, but adverse drug event (ADE) prediction using EHR pretraining models is a field much left to explore. Tertiary hospitals have good-quality inpatient records since all medical events that occur in the hospital are recorded in the EHR database. Therefore, EHR data from tertiary hospitals are well suited for monitoring the condition of hospitalized patients, especially the risk of ADE.

[Revised] EHR pretraining models exhibited great performance in various medical tasks, but adverse drug event (ADE) prediction using EHR pretraining models is a field much left to explore. Tertiary hospitals have good-quality inpatient records since all medical events that occur in the hospital are recorded in the EHR database. For instance, lab or vital signs variation before and after medication intake and condition and procedure as a coping strategy for ADE, which are stored in EHR, can reflect the topical or systemic impact of numerous drugs for various types of patients. Therefore, EHR data from tertiary hospitals are well suited for monitoring the condition of hospitalized patients, especially the risk of ADE.

The last paragraph of **Introduction**.

[Before] In this study, we exhibited the validity of EHR pretraining model in ADE prediction tasks using CDM-based EHR data, which includes various domains such as diagnosis, prescription, measurement, and procedure. For pretraining, we adopted the masked language model (MLM). (...)

[Revised] In this study, we exhibited the validity of EHR pretraining model in ADE prediction tasks using EHR data based on an observational medical outcomes partnership (OMOP)-common data model (CDM). OMOP-CDM (hereafter CDM) enables the transformation of different databases to a standardized format, facilitating multicenter study with different EHR databases and external validation of statistical models or artificial intelligence methodologies for EHR data analysis. We used enormous records of diagnosis, prescription, measurement, and procedure domains in CDM from two separate tertiary hospitals in Korea. For pretraining, we adopted the masked language model (MLM). (...)

Comment 2:

The review of current studies is insufficient. There are several BERT-based and transformer-based EHR pre-trained model functions in clinical domains. Reviewing the current studies and pointing out the empty in ADE prediction would be better. Also, studies in ADE prediction using other methods are preferred to summarize their disadvantages.

Answer:

Thank you for your valuable comment. The second paragraph of the introduction describes that deep learning-based EHR analysis models and learning patient representation through pretraining has become an area ripe for exploration. In addition, we explained the mechanism of language model-based EHR pretraining models (review of current studies). However, readers might regard this explanation not enough, so we have added the examples of some EHR pretraining models.

For pointing out the empty in ADE prediction using EHR pretraining models, we mentioned that EHR pretraining models had hardly applied in ADE prediction at the beginning of the third paragraph in introduction. However, as the reviewer suggested, we acknowledge that summarizing ADE prediction using other methods would improve our manuscript. Accordingly, we have revised our manuscript with the description of machine learning-based ADE prediction studies. Thank you again for your helpful opinion to enhance the manuscript.

Changes made in the manuscript:

The second paragraph of **Introduction**.

[Before] (...) For pretraining tasks, most of those prior studies utilized bidirectional encoder representation from transformers (BERT), one of the most famous models for contextualizing sequential inputs, especially natural language. (...)

[Revised] (...) For pretraining tasks, most of those prior studies (e.g., BEHRT, Med-BERT, and CEHR-BERT) utilized bidirectional encoder representation from transformers (BERT), one of the most famous models for contextualizing sequential inputs, especially natural language. (...)

The third paragraph of **Introduction**.

[Before] (...) Therefore, EHR data from tertiary hospitals are well suited for monitoring the status of hospitalized patients, especially the risk of ADE.

[Revised] (...) Therefore, EHR provides new opportunities to address limitations of traditional approaches and promote discovery-oriented pharmacovigilance.¹ Previous studies utilized odds ratio, regression-based modeling (e.g., logistic, Poisson, and Cox regression), and tree-based machine learning models for EHR-based ADE management, but no study had utilized EHR pretraining models for ADE prediction.

Comment 3:

The prediction task set a timepoint 4 weeks before ADE as the cohort requires an ADE within 4 weeks of prescription. As illustrated in Fig 1b, the target drug and related ADE may both occur after the prediction timepoint. How to explain the prediction task without a prescription at the time of prediction timepoint? Since 2,4,8,12 weeks' timepoint may cause the prescription to occur before or after the prediction timepoint, will the task setting influence the performance of the validation of different prediction timepoints?

Answer:

Thank you for your valuable comment. In this study, the prediction timepoint was set to 4 weeks before the occurrence of an ADE. If the prediction timepoint is set to a prescription and the ADE occurs after a short time, it is much easier to predict ADE because there could be a lot of information directly related to the target ADE. This study tried to exhibit the power of EHR pretraining models in predicting various ADEs, and we applied the same cohort definition for those various ADEs roughly. So, we focused more on removing potential target leakage (input data containing information closely related to the target ADE) by setting the prediction timepoint far from the ADE occurrence than a prescription at the time of the prediction timepoint.

As the reviewer commented, the prescription can exist before or after the prediction timepoint, 2, 4, 8, 12 weeks' timepoint can cause the prescription occur before or after the prediction timepoint, and the different prediction timepoints can influence the prediction performance. The different prediction timepoints were considered to address the potential concern that each drug may have a different duration of action. All patients in case groups were prescribed target drug between the prediction timepoint and the ADE occurrence. The amount of information according to the prediction timepoint was our main concern, instead of whether specific prescriptions were included or not.

We acknowledge that the reviewer's interest may arise by readers and have revised the manuscript with additional explanations accordingly.

Changes made in the manuscript:

“ADE prediction task” section in Methods.

[Before] (...) For the case group, we set a prediction timepoint to four weeks before the adverse event occurrence. (...)

[Revised] (...) For the case group, we set a prediction timepoint to four weeks before the adverse event occurrence to prevent target leakage, assuming that no obvious symptom records directly related to the target ADE exist prior to this timepoint. (...)

[Before] (...) Given that adverse events can vary in timing for different drugs and circumstances, we conducted additional experiments by setting the prediction timepoints to two, eight, and twelve weeks before the occurrence of adverse events.

[Revised] (...) Given that the duration of action can vary for different drugs and circumstances, we conducted additional experiments by setting the prediction timepoints to two, eight, and twelve weeks before the occurrence of adverse events. In addition, diverse prediction timepoints were to consider impact of the different amount of information.

The fifth paragraph of **Discussion**.

[Before] First, we defined cohorts very roughly, even though several critical exclusion criteria might exist for proper cohort analysis.

[Revised] First, we defined cohorts very roughly, even though several critical exclusion criteria might exist for proper cohort analysis. This approach was to demonstrate the effectiveness of EHR pretraining models in predicting various ADEs. For future work, we suggest conducting cohort studies with more rigorously defined cohort criteria and detailed analysis using our proposed methods.

Comment 4:

Fig 3 and Fig 4 showed the feature importance of 3 ADEs. As there are 7000/41536 common concepts in both hospitals, what are the common features and unique features of the top 10s?

Answer:

For the anticoagulant-intracranial hemorrhage cohorts, aPTT, triglyceride, Holter recording, Radiography of nasal sinuses, and Chest pain were the most important common features. The drug codes were not included because SNUH and AUMC used different concept IDs. In detail, both hospitals regarded aspirin as important; however, the exact concept IDs and corresponding concept names were different. As shown in the revised Supplementary Table 3 in Comment 5 of Reviewer 2, many tokens were not shared between the two hospitals (only around 4% for prescription). Thus, important common concepts are not representative, and those concepts can cause misunderstandings among readers. Instead, model interpretations were validated qualitatively by the clinicians in our study.

Comment 5:

Table 2 indicated an unbalanced data distribution of the cohort where the control group is much larger than the case group. Still, the model presented a good recall. There should be a comparatively large false positive, though it is acceptable to catch any possible ADE. It is better to report the percentage of case and control in the study cohort characteristics. Discussing examples of false positives and negatives is preferred to better explain the validation result.

Answer:

Thank you for your insightful comment. Patient demographics and incidence ratio of each ADE cohort are summarized in Supplementary Table 4. We exhibited the example for feature importance of false positives and negatives in Fig. R1 on the next page. Patients in the false positive group had no ADE, but the calculated risk was high. Thus, we can infer that those patients might be more hazardous with the ADE despite its absence.

As the reviewer commented, analyzing false positives (negatives) can attract readers' interest. However, since the cohorts in this study were broadly defined and our primary aim was to demonstrate the effectiveness of EHR pretraining models for predicting various ADEs, we are concerned that such analysis might distract from the study's focus and lead to potential misunderstandings regarding ADE-related knowledge. We believe that more detailed analysis would be better suited to further clinical studies with more rigorously defined cohorts. Accordingly, we have revised our manuscript suggesting detailed clinical investigation with our proposed method.

Changes made in the manuscript:

The fifth paragraph of **Discussion**.

[Before] First, we defined cohorts very roughly, even though several critical exclusion criteria might exist for proper cohort analysis.

[Revised] First, we defined cohorts very roughly, even though several critical exclusion criteria might exist for proper cohort analysis. This approach was to demonstrate the effectiveness of EHR pretraining models in predicting various ADEs. For future work, we suggest conducting cohort studies with more rigorously defined cohort criteria and detailed analysis using our proposed methods.

Specific comments: Reviewer #4

Comment 1:

I would like to see more discussion around some of the features and what their relevance is for future implementation work. Many of the features are secondary to known risk factors for these adverse events (ie as you describe ISMN use would strongly suggest the patient has CVS disease, speech audiometry likely to be strongly correlated with advanced age). If the model has simply identified known and well established risk factors that a clinician uses in their assessment how would this add benefit in a clinical workflow? This will not be immediately obvious to clinicians reading the paper at present.

Answer:

Thank you for your valuable comment. Since our proposed methods took account of all patient records without any exceptions, there were indirectly related features inevitably. However, as the reviewer commented, it would be better if the model could present well-established risk factors related to clinicians for each patient. Patient-level interpretation is available (Supplementary Fig. 2), and thus, presenting comprehensible explanation for patients with ADE risk using our proposed method would be promising future work. Accordingly, we have revised the manuscript suggesting the future work.

Changes made in the manuscript:

The fifth paragraph of **Discussion**.

[Before] Second, there were irrelevant important features for each cohort. For example, speech audiometry, which is extraneous with all the ADE in this study, was found in the NSAID-PU and AC-ICH cohorts. It might be associated with overfitting, black box, and age, as PU and ICH usually occur in aged patients, but further investigation is needed.

[Revised] Second, there were irrelevant important features for each cohort. For example, speech audiometry, which is extraneous with all the ADE in this study, was found in the NSAID-PU and AC-ICH cohorts. It might be associated with overfitting, black box, and age, as PU and ICH usually occur in aged patients, but further investigation is needed. Additionally, for future work, we suggest leveraging large language models (LLMs) to provide well-established explanations with identified known risk factors for each patient at ADE risk, making the results more interpretable for clinicians.

Comment 2:

The paper would be further enriched by expanding on the limitations - particularly the fact that the work was completed on two large, well constructed and OMOP mapped datasets, with high quality EHR data, which is not the case in a large percentage of the world, limiting external validity. Further validation work on external datasets Is essential (I appreciate this is likely to be in a future paper)

Answer:

Thank you for your valuable comment. According to the reviewer's comment, we have revised the limitation suggesting further validation work on global external datasets.

Changes made in the manuscript:

The fifth paragraph of **Discussion**.

[Before] Fourth, our model was only trained by structured EHR data. For future work, utilizing several types of unstructured medical data and combining those data types are suggested.

[Revised] Fourth, our model was only trained by structured EHR data from a single nation. For future work, we suggest utilizing and combining various types of unstructured medical data, as well as conducting external validation using global datasets.

References

- 1 Davis, S. E. *et al.* Use of Electronic Health Record Data for Drug Safety Signal Identification: A Scoping Review. *Drug Safety* **46**, 725-742 (2023). <https://doi.org:10.1007/s40264-023-01325-0>